Machine Learning I – Exam Prep Q&A;

Digitization and Data Analytics · TU Dresden (CIDS/ZIH) · Based on lecture: "Digitization and Data Analytics – Machine Learning I" (J. Klimpke)

This document collects likely exam-style questions for this lecture, broken into the smallest sub-question units (as typically asked in TU Dresden DDA past exams), each followed by a concise model answer. Organized by lecture section: (1) Introduction to Machine Learning, (2) Similarity and Distances, (3) Clustering — Overview, Partitioning, Hierarchical, Density-based, Cluster Assessment, Big Data Aspects.

1. Introduction to Machine Learning

1.1 Data Processing & Analytics Pipeline

Q: Name and briefly describe the four stages of the Data Processing and Analytics cycle.

A: The cycle is a closed loop of four stages:

- Data Collection – gathering raw data from the source.
- Data Preprocessing – Extraction, Reduction, Annotation, Aggregation, Cleaning.
- Data Analysis – Prediction, Exploration, Modeling (this is where Machine Learning is applied).
- Interpretation – making sense of analysis results, which feeds back into further data collection.

Q: Where in the Data Processing and Analytics cycle is Machine Learning positioned?

A: Machine Learning provides the automated methods used in the Data Analysis step (Prediction, Exploration, Modeling), which follows Data Preprocessing and precedes Interpretation.

1.2 Definition & Goals of Data Analytics / Machine Learning

Q: What are the two goals of Data Analytics?

A: Recognize complex potential patterns in existing data.

- Support decision making.

Q: Give Murphy's (2012) formal definition of Machine Learning.

A: "We define machine learning as a set of methods that can automatically detect patterns in data, and then use the uncovered patterns to predict future data, or to perform other kinds of decision making under uncertainty." (Murphy, K.P., Machine Learning – A Probabilistic Perspective, MIT Press, 2012)

Q: Why does big data analytics require Machine Learning?

A: Because big data analytics needs automated methods, and Machine Learning provides exactly such automated methods of data analysis (manual/statistical analysis alone does not scale).

1.3 Types of Machine Learning

Q: List the four main types of Machine Learning.

A: Supervised Learning (Predictive Analysis)

- Unsupervised Learning (Descriptive/Exploratory Analysis)
- Semi-Supervised Learning
- Reinforcement Learning

Q: Describe Supervised Learning informally: what question does it answer and what is its goal?

A: It describes properties of future data; answers the question "What could happen?"; goal is predicting the future, e.g., forecasting, scoring, classification of unknown data.

Q: Give the formal definition of Supervised Learning (classification): what is given, and what is the goal?

A: Given a training set D = {(x_j, y_j)} for j=1..N of N labeled input-output pairs, where x is the feature vector (attributes/covariates/predictors) and y is the output vector (label/response/target).

- Goal: learn a mapping f: x -> y.

Q: What is the probabilistic view of supervised learning?

A: Fitting a conditional model p(y|x), which specifies a distribution over outputs y given inputs x.

Q: Describe Unsupervised Learning informally: what question does it answer and what is its goal?

A: It describes existing data; answers the question "What has happened?"; goal is to extract potential patterns (e.g., discover groups within data).

Q: Give the formal definition of Unsupervised Learning: what is given, and what is the goal?

A: Given a training set D = {x_j} for j=1..N of N unlabeled inputs (no y).

- Goal: find interesting patterns, e.g., group data points x_j based on an application-dependent distance measure.

Q: What is the probabilistic view of unsupervised learning, and what is its key challenge?

A: Fitting an unconditional model p(x), which can generate new data x.

- Challenge: deciding what kind of pattern is interesting/should be searched for, and how to assess the extracted pattern.

Q: Using the Iris flower example, explain the key difference between supervised and unsupervised learning outcomes.

A: In supervised learning the data is labeled with the species (setosa/versicolor/virginica) and the goal is to classify unknown flowers into these known classes (e.g., via a decision tree using Petal Length/Width).

- In unsupervised learning (e.g., k-means) the data is unlabeled; the algorithm discovers groups (clusters) purely from feature similarity, and the learned model does NOT include the semantic meaning of the clusters — the clusters found need not correspond exactly to the true species.

Q: What are Semi-Supervised Learning and Reinforcement Learning, and are they covered in this lecture series?

A: Semi-Supervised Learning uses both labeled and unlabeled data.

- Reinforcement Learning learns via occasional reward/punishment signals (e.g., a baby learning to walk, dog training).
- Both are NOT topics of this lecture series (mentioned only for completeness).

1.4 Machine Learning Takeaway Diagram

Q: Complete the taxonomy: Machine Learning splits by data type into which two branches, and each of those splits further into which two categories?

A: Supervised Learning (uses labeled data) splits by data type into: Classification (discrete data) and Regression (continuous data).

- Unsupervised Learning (uses unlabeled data) splits into: Clustering and Dimensionality Reduction.

Q: Name two typical algorithms for each of: Classification, Regression, Clustering, Dimensionality Reduction.

A: Classification: Decision Tree, Naïve Bayes Classifier, Random Forest, Support Vector Machines, Artificial Neural Networks.

- Regression: Decision Tree, Linear Regression, Artificial Neural Networks.
- Clustering: Partitioning, Hierarchical Clustering.
- Dimensionality Reduction: PCA (Principal Component Analysis).

2. Similarity and Distances

2.1 Foundations

Q: Give the formal statement of a similarity/distance problem.

A: Given two objects O1 and O2, determine a value of the similarity Sim(O1,O2) (or distance Dist(O1,O2)) between the two objects.

Q: List possible object types for which similarity/distance can be computed.

A: Images, Text Documents, Time Series, Mixed data, (Graphs, Categorical/Sequence data), etc.

Q: Explain, using the alphabet example, why "green" and "blue" are considered more similar than "green" and "red".

A: The example encodes similarity via positional/character distance in a sequence (e.g., letters of the alphabet). It illustrates that similarity is not absolute — it depends on how it is measured/defined for the given representation, motivating the need for a well-chosen distance function.

Q: Quote Aggarwal's key statement about the importance of choosing a distance function.

A: "Poor choices of the distance function can sometimes be disastrously misleading depending on the application domain." (Aggarwal, C., Data Mining – The Textbook)

Q: Define Similarity and Dissimilarity (distance), including typical value ranges.

A: Similarity: a numerical measure of how alike two objects are; higher value = more alike; often in range [0,1].

- Dissimilarity (distance): a numerical measure of how different two objects are; lower value = more alike; minimum dissimilarity is often 0; upper limit varies (not fixed).

2.2 Distances for Multidimensional (Quantitative) Data

Q: Give the general formula of the L_p-norm (Minkowski distance) for two vectors x, y in R^d.

A: Dist(x,y) = (sum_{i=1}^{d} |x_i - y_i|^p)^(1/p)

Q: Which distance is obtained for p=1, p=2, and p=infinity? Name each.

A: p = 1: L1-norm = Manhattan distance (sum of absolute differences).

- p = 2: L2-norm = Euclidean distance (square root of sum of squared differences).
- p = infinity: L_infinity-norm = Maximum (Chebyshev) distance = max_i |x_i - y_i|.

Q: Worked example: Given x1=(1,2), x2=(3,5), x3=(2,0), x4=(4,5), compute the Manhattan (L1), Euclidean (L2), and Maximum (L-infinity) distance between x1 and x2.

A: Differences: |1-3|=2, |2-5|=3.

- Manhattan: 2+3 = 5.
- Euclidean: sqrt(2^2+3^2) = sqrt(4+9) = sqrt(13) ≈ 3.61.
- Maximum: max(2,3) = 3.
- (These match the lecture's distance matrices: d(x1,x2) = 5, 3.61, 3 respectively.)

2.3 Similarity for Categorical Data

Q: Why can't standard L_p-norms be used directly for categorical data?

A: Because there is no ordering among the discrete values of categorical attributes, so numerical differences x_i - y_i are not meaningful; similarity must instead be based on whether attribute values match.

Q: Give the simplest per-attribute similarity function S(x_i, y_i) for categorical data, and the total similarity formula.

A: S(x_i, y_i) = 1 if x_i = x_j (match), 0 otherwise (mismatch).

- Total similarity: S(x, y) = sum_{i=1}^{d} S(x_i, y_i).

Q: Define the Inverse Occurrence Frequency approach for categorical similarity, including the formula for p_k(x) and S(x_i, y_i).

A: p_k(x) = (number of datapoints where the k-th attribute equals x) / (total number of datapoints).

- S(x_i, y_i) = 1 / p_k(x_i)^2 if x_i = x_j, and 0 if x_i ≠ x_j.
- Intuition: rarer matching values contribute a higher similarity score than common ones.

Q: Worked example: In a table of 6 people, the "Adult" attribute has values {1,1,1,0,1,0}, giving p_Adult(0)=1/3 and p_Adult(1)=2/3. What is S(x_i,y_i) for two people who are both Adult=1, using inverse occurrence frequency?

A: S = 1 / p_Adult(1)^2 = 1 / (2/3)^2 = 1 / (4/9) = 9/4 = 2.25.

2.4 Similarity for Text Data

Q: Give the Cosine Similarity formula for two document vectors x, y, and state two key properties.

A: cos(x,y) = (sum_{i=1}^{d} x_i·y_i) / (sqrt(sum x_i^2) · sqrt(sum y_i^2)).

- Property 1: it is insensitive to the absolute length of the document (measures angle, not magnitude).
- Property 2: it simply uses raw (term) frequencies as the vector components.

Q: Name three application domains for cosine similarity.

A: Information retrieval, biologic taxonomy, gene feature mapping.

2.5 Similarity for Discrete Sequences

Q: What type of data does Edit Distance (Levenshtein distance) apply to, and how is it defined?

A: Applies to sequences of equal length with categorical values.

- Edit distance = the least amount of cost required to transform one sequence into another, using insertions, deletions, and replacements, each with a specific cost; the total distance depends on the costs chosen.

Q: Worked example: transforming 'abababab ab' into 'bababababa' — compare the cost using (a) only replacements vs (b) using one deletion + one insertion.

A: (a) Replacing every character (a<->b swap) costs 10 replacements.

- (b) Deleting the leading 'a' and inserting an 'a' at the right end costs only 1 deletion + 1 insertion = 2 operations — far cheaper, showing the choice of operations matters.

Q: Define Longest Common Subsequence (LCSS) and state its computational complexity.

A: LCSS is the longest subsequence common to two sequences (elements need not be contiguous, but must keep relative order); e.g., for 'xmjyauz' and 'mzjawxu' the LCSS is 'mjau'.

- LCSS can be computed in polynomial time (PTime), e.g., via dynamic programming.

2.6 Similarity for Graphs

Q: Distinguish between similarity between two nodes and similarity between two whole graphs.

A: Between nodes: based on connectivity — via Shortest paths (e.g., Dijkstra's algorithm) or via Many paths (e.g., PageRank, SimRank).

- Between two graphs: much more challenging — the graph-isomorphism problem (checking if two graphs are identical) is NP-hard.

Q: Name three approaches used to define similarity/distance between two graphs.

A: Maximum common subgraph distance, substructure-based similarity, graph-edit distance.

2.7 Similarity for Time Series

Q: List the four problems that make plain L_p-norm distances unsuitable for comparing raw time series.

A: Time series are not drawn on the same scale (behavioral attribute scaling/translation).

- Absolute time stamp is not important — series may need to be shifted (temporal attribute translation).
- Time series have noisy segments.
- The scale of the temporal axis may differ (temporal attribute scaling).

Q: What is Dynamic Time Warping (DTW) and how does it solve the above problems?

A: DTW stretches a series along the time axis and repeats some elements of either series so that both series end up with the same length; once aligned/warped to equal length, any standard distance measure (e.g., Euclidean) can then be applied to compare them.

2.8 High-Dimensional Data & Curse of Dimensionality

Q: What characterizes high-dimensional data, and give one example application area.

A: $n << d$: the number of observations n is much smaller than the number of dimensions d (e.g., $n=100$ observations in $d=10^6$ dimensions).

- Example applications: genome analysis, spatial statistics.

Q: Describe the "Curse of Dimensionality" — list its four key symptoms mentioned in the lecture.

A: The center of high-dimensional spaces is almost empty.

- Points get scattered to the corners and boundary regions.
- The majority of vectors are (almost) orthogonal to each other.
- Projection onto a lower-dimensional space may cause severe loss of information.

Q: According to Leskovec/Rajaraman/Ullman, what happens to pairwise distances in high-dimensional spaces?

A: Almost all pairs of points end up at about the same distance from one another, and almost any two vectors are almost orthogonal — this is why clustering that 'looks easy' in 2D becomes hard in 10 or 10,000 dimensions.

2.9 Similarity and Distances — Takeaway

Q: Summarize the four key takeaway points about similarity and distance measures.

A: The choice of similarity/distance measure is essential for successful ML applications and can lead to very different results.

- Similarity/distance measures are data dependent.
- Similarity/distance measures are application dependent.
- High-dimensional data behave differently (curse of dimensionality).

3. Clustering — Overview

Q: Define a Cluster and define Clustering (cluster analysis / data segmentation).

A: Cluster: a collection of data objects that are similar (related) to one another within the same group, and dissimilar (unrelated) to objects in other groups.

- Clustering: finding similarities between data according to characteristics found in the data, and grouping similar data objects into clusters.

Q: Name the two general application contexts for clustering.

A: As a stand-alone tool to get insight into data distribution.

- As a preprocessing step for other algorithms (e.g., outlier detection).

Q: Give at least four application domains for clustering discussed in the lecture, with an example use case for each.

A: Biology — taxonomy of living things (kingdom, phylum, class, order, family, genus, species).

- Information retrieval — document clustering.
- Land use — identifying areas of similar land use in earth observation data.
- Marketing — discovering distinct customer groups for targeted marketing.
- City-planning — grouping houses by type, value, and location.
- Climate — finding atmospheric/ocean patterns.
- Economic Science — market research.

3.1 Considerations for Clustering

Q: List the four key considerations/criteria when designing or choosing a clustering approach, with an example contrast for each.

A: Partitioning criteria: single-level vs. hierarchical partitioning (multi-level hierarchical partitioning is often desirable).

- Separation of clusters: exclusive (e.g., one customer belongs to only one region) vs. non-exclusive (e.g., one document may belong to more than one class).
- Similarity measure: distance-based (e.g., Euclidean, road network, vector) vs. connectivity-based (e.g., density or contiguity).
- Clustering space: full space (often for low-dimensional data) vs. subspaces (often used in high-dimensional clustering).

Q: What precondition must hold for clustering to make sense, and what two properties define a high-quality clustering?

A: Precondition: existing patterns must be present within the data (the application should give hints supporting this hypothesis).

- High intra-class similarity: clusters are cohesive internally.
- Low inter-class similarity: clusters are distinctive from one another.

Q: What three factors does the quality of a clustering method depend on?

A: The similarity measure used by the method.

- Its implementation.
- Its ability to discover some or all of the hidden patterns.

Q: Why is clustering considered a hard problem, especially in high dimensions?

A: Clustering in two dimensions and with small amounts of data looks (and is) easy, but many real applications involve 10 or even 10,000 dimensions.

- In high-dimensional spaces: almost all pairs of points are at about the same distance, and almost any two vectors are almost orthogonal — the "Curse of Dimensionality" (Leskovec, Rajaraman, Ullman — Mining of Massive Datasets).

3.2 Major Clustering Approaches

Q: Name the four major clustering approach families and one typical algorithm for each.

A: Partitioning approach — construct various partitions, evaluate by a criterion (e.g., minimize sum of squared errors); typical methods: k-means, k-medoids, CLARANS.

- Hierarchical approach — create a hierarchical decomposition using some criterion; typical methods: DIANA, AGNES, BIRCH, CAMELEON.
- Density-based approach — based on connectivity and density functions; typical methods: DBSCAN, OPTICS, DenClue.
- Grid-based approach — based on a multiple-level granularity structure; typical methods: STING, WaveCluster, CLIQUE.

4. Clustering — Partitioning (k-Means / k-Medoids)

Q: Formally define the partitioning problem: what is X, what are C_i, and what is minimized?

A: A set X = {x1,...,xn} of n objects is partitioned into a set of k clusters C_i, i = {1,...,k}.

- Distance is measured to the centroid or medoid c_i of cluster C_i.
- The sum of squared distances E = sum_{i=1}^{k} sum_{p in C_i} dist(p, c_i)^2 is minimized.

Q: Give the formulas for centroid and medoid of a cluster.

A: centroid = (1/n) · sum_{i=1}^{n} x_i (the mean point of the cluster).

- medoid = argmin_{y in X} sum_{i=1}^{n} dist(y, x_i) (the object in the cluster minimizing total distance to all other members — i.e., the most centrally located actual data point).

Q: Contrast the two partitioning criteria: global optimal vs. heuristic methods. Which two heuristic algorithms are named, and how does each represent a cluster?

A: Global optimal: exhaustively enumerate all possible partitions (computationally infeasible for realistic data sizes).

- Heuristic methods are used instead:
- k-means (MacQueen 1967; Lloyd 1957/1982): each cluster is represented by the center (mean) of the cluster.
- k-medoids / PAM — Partition Around Medoids (Kaufman & Rousseeuw 1987): each cluster is represented by one of the actual objects in the cluster (the medoid).

Q: Describe the k-Means algorithm as a step-by-step procedure.

A: 1) Partition objects into k non-empty subsets (e.g., arbitrarily assign objects to k groups, or pick k initial centroids).

- 2) Repeat: (a) Compute the centroid (mean point) for each partition; (b) Assign each object to the cluster of its nearest centroid.
- 3) Until no change occurs (convergence).

Q: Who is credited with the k-means algorithm, and what is the reference citation given in the lecture?

A: Lloyd, Stuart P. "Least squares quantization in PCM." IEEE Transactions on Information Theory 28.2 (1982): 129-137 (also attributed to MacQueen, 1967).

Q: What is the main weakness of the k-means algorithm, and why does it occur?

A: k-means is sensitive to outliers: an object with an extremely large (atypical) value can substantially distort the mean, and thus distort the distribution/position of the cluster centroids.

Q: How does k-medoids address the weakness of k-means, and what statistical concept is exploited?

A: Instead of taking the mean value of the objects in a cluster as the reference point (which is sensitive to outliers), k-medoids uses the medoid — the most centrally located ACTUAL object in the cluster — as the reference point. Because it is an actual data point (like a median rather than a mean), it is far less sensitive to extreme outlier values.

5. Clustering — Hierarchical Clustering

Q: What is the main idea of hierarchical clustering, and what two key characteristics distinguish it from partitioning approaches?

A: Main idea: define a distance between clusters, then merge/split clusters based on that distance, using a distance matrix as the clustering criterion.

- Characteristic 1: hierarchical clustering does NOT require the number of clusters k as an input.
- Characteristic 2: hierarchical clustering DOES need a termination condition.

Q: Distinguish the two directions of hierarchical clustering: agglomerative and divisive. Name the algorithm associated with each.

A: Agglomerative (AGNES – AGglomerative NESting): bottom-up — starts with each object as its own cluster and successively merges the closest clusters until all belong to one cluster.

- Divisive (DIANA – DIvisive ANAlysis): top-down — starts with all objects in one cluster and successively splits clusters until each object forms its own cluster (the inverse order of AGNES).

Q: Give the definitions (formulas) of Single linkage, Complete linkage, and Average linkage distance between two clusters G and H.

A: Single linkage: $d_{SL}(G,H) = \min_{x \in G, y \in H} d(x,y)$ — distance between the two closest members.

- Complete linkage: $d_{CL}(G,H) = \max_{x \in G, y \in H} d(x,y)$ — distance between the two most distant members.
- Average linkage: $d_{Avg}(G,H) = (1 / (n_G \cdot n_H)) \cdot \sum_{x \in G} \sum_{y \in H} d(x,y)$, where $n_G=|G|$, $n_H=|H|$ — average pairwise distance between all points of the two clusters.

Q: Describe AGNES step by step, and name its introducing authors/year.

A: Introduced 1990 by Kaufman, L. and Rousseeuw, P.

- AGNES uses the single-link method and the dissimilarity matrix: repeatedly merge the two nodes (clusters) that have the least dissimilarity, until eventually all nodes belong to the same cluster.
- Implemented in statistical packages such as R.

Q: Describe DIANA and explain its relationship to AGNES.

A: Introduced 1990 by Kaufman, L. and Rousseeuw, P.

- DIANA works in the inverse order of AGNES: it starts from one cluster containing all points and recursively splits clusters until eventually each node forms a cluster of its own.
- Also implemented in statistical packages such as R.

Q: Based on the Iris dendrogram comparison in the lecture, how does the choice of linkage method (single vs. complete vs. average) affect the resulting cluster structure for the same distance measure (Euclidean)?

A: Different linkage methods produce different dendrogram structures and different implied groupings, even though the same underlying distance (Euclidean) is used — e.g., single linkage tends to chain points together (elongated clusters), while complete and average linkage tend to produce more compact, balanced clusters. This shows that BOTH the distance measure AND the linkage method must be specified and both influence the result.

Q: List the Pros and Cons of hierarchical clustering.

A: Pros: flexibility due to the use of complex distance measures; no other parameters than the distance function and the fusion (linkage) method; the cluster result allows substructures (a full hierarchy of nested clusters).

- Cons: analysis effort of the result is high; cluster analysis provides MANY partitions of the data (one per cut level of the dendrogram); the user must decide how/where to partition (cut the tree).

Q: What is the general time complexity of hierarchical clustering, and to what can it be reduced?

A: In general: $O(N^3)$ time, with N = number of data points.

- Can be reduced to $O(N^2 \cdot \log(N))$ time (using appropriate data structures).
- Special case — single link: can be computed in $O(N^2)$ time.

6. Clustering — Density-Based (DBSCAN)

Q: What does DBSCAN stand for, and who introduced it (citation)?

A: DBSCAN = Density-Based Spatial Clustering of Applications with Noise.

- M. Ester, H.-P. Kriegel, J. Sander, X. Xu. "A density-based algorithm for discovering clusters in large spatial databases with noise." KDD'96, AAAI Press, 226-231.

Q: Name the two parameters of DBSCAN and define each.

A: ε (epsilon): the maximum distance within which points are considered reachable from one another.

- minPts: the minimum number of points required within the ε-distance (neighborhood) to define a core point.

Q: Define the three types of points in DBSCAN: Core, Border, and Noise.

A: Core point: has at least minPts points (including itself) within distance ε.

- Border point: is connected/reachable from a core point, but itself has fewer than minPts points within distance ε.
- Noise point: is neither a core point nor a border point.

Q: Outline the DBSCAN algorithm (pseudocode-level) in your own words.

A: For each unvisited point P in the dataset: mark P visited; find its ε-neighborhood via regionQuery(P, eps).

- If the neighborhood has fewer than minPts points, mark P as NOISE.
- Otherwise, start a new cluster C and call expandCluster(P, NeighborPts, C, eps, MinPts), which adds P to C and recursively adds all density-reachable neighbors (re-classifying any NOISE points found along the way as border points of C if they connect to a core point).
- regionQuery(P, eps) simply returns all points within P's ε-neighborhood (including P itself).

Q: Compare DBSCAN and k-Means on datasets with non-convex cluster shapes (e.g., two concentric rings, or two interleaved half-moons). Which handles these better, and why?

A: DBSCAN handles them correctly because it groups points by density-connectivity rather than by proximity to a single center, so it can identify arbitrarily shaped (non-convex) clusters and can also isolate noise/outlier points as their own (small) group.

- k-Means fails on such shapes because it partitions space via distance to centroids, effectively cutting clusters with straight (Voronoi) boundaries through the middle of a ring or moon shape, regardless of the true underlying density structure.

7. Clustering — Cluster Assessment

7.1 Assessing Clustering Tendency

Q: What is the purpose of assessing clustering tendency, and what statistical test is used?

A: Purpose: to assess whether a non-random structure actually exists in the data before clustering it — i.e., to estimate the probability that the data was generated by a uniform (random) distribution.

- Test used: the Hopkins Statistic (Cross, G.R.; Jain, A.K., 1982, "Measurement of clustering tendency"), which tests spatial randomness.

7.2 Determining the Number of Clusters

Q: Describe the Elbow Method for choosing k, and state its main drawback.

A: Compute the sum of squared distances (within-cluster sum of squares) for each candidate k.

- The optimal number of clusters is indicated by a sharp turn ("elbow") in the resulting graph, after which adding more clusters yields diminishing improvement.
- Drawback: it is very subjective, as it is often ambiguous exactly where the elbow is located.

Q: Define the Silhouette Coefficient sc(i) for a data point i, including the meaning of a_i and b_i, and its value range.

A: $sc(i) = (b_i - a_i) / \max(a_i, b_i)$.

- a_i: mean distance from point i to the other points in the SAME cluster — a measure of compactness.
- b_i: mean distance from point i to points in the NEXT closest (different) cluster — a measure of separation.
- Range: $-1 \leq sc(i) \leq 1$ (values near +1 indicate good clustering, near 0 indicate overlapping clusters, negative values indicate likely misassignment).

Q: How is the Silhouette Score of an entire clustering K defined, and how would you use it to select k?

A: The silhouette score of clustering K is defined as the mean silhouette coefficient over all instances/data points.

- To select k, compute the silhouette score for several candidate values of k and choose the k that maximizes the (mean) silhouette score.

7.3 External / Manual / Indirect Evaluation

Q: What is required for External evaluation of a clustering, and name three concrete metrics used.

A: Requires "ground truth" — i.e., known true labels for the data (e.g., the species labels in the Iris dataset).

- Metrics: Purity; Jaccard Index $J(A,B) = |A \cap B| / |A \cup B|$; F-score / Confusion Matrix (as also used in supervised learning evaluation).

Q: Describe Manual evaluation and Indirect evaluation of clustering results.

A: Manual evaluation: assessing the clustering result by human judgment/inspection.

- Indirect evaluation: assessing the clustering indirectly by evaluating the performance of the downstream application that uses the clustering result.

8. Clustering — Aspects of Big Data Clustering

Q: What is the central challenge of clustering big data, and give a concrete example of algorithmic complexity that motivates it.

A: Central challenge: Scalability — clustering algorithms often have high time complexity (e.g., the k-means optimization problem is NP-hard in general), so they must be sped up / scaled up while sacrificing as little clustering quality as possible.

Q: Describe the generic two-phase procedure most clustering algorithms follow (relevant for scaling).

A: Initialization phase: e.g., randomly choose k initial cluster centers.

- Iterative phase: repeat an iterative process until some convergence criterion is met (e.g., in k-means: assign each data point to its closest cluster center, then update centers, and repeat).

Q: Name three general strategies to speed up / scale up clustering for big data.

A: Reducing the number of iterations needed.

- Reducing access to data points, e.g., via randomized/sampling techniques.
- Distributing / parallelizing the computation (e.g., across a compute cluster).

9. Full-Lecture Summary Recap (Quick-Fire Q&A;)

Q: In one sentence each, distinguish: Classification, Regression, Clustering, Dimensionality Reduction.

A: Classification: supervised learning of a mapping to DISCRETE labels.

- Regression: supervised learning of a mapping to CONTINUOUS values.
- Clustering: unsupervised discovery of groups (discrete structure) in unlabeled data.
- Dimensionality Reduction: unsupervised discovery of a lower-dimensional (continuous) representation of unlabeled data, e.g., PCA.

Q: List the three clustering algorithm families explicitly worked through in this lecture, with one algorithm each.

A: Partitioning — K-Means (and K-Medoids/PAM).

- Hierarchical clustering — AGNES / DIANA.
- Density-based — DBSCAN.

Q: What does clustering fundamentally depend on, and why is assessing a clustering result described as "a challenge"?

A: Clustering fundamentally depends on the chosen distance/similarity measure — different choices can produce very different clusterings.

- Assessment is a challenge because, without ground truth, quality must be judged indirectly (e.g., Hopkins statistic, Elbow method, Silhouette method); with ground truth, external metrics like Purity, Jaccard Index, or F-score can be used instead.

Q: What three aspects make clustering LARGE datasets particularly difficult?

A: The need to diminish/sample the set of data processed at once.

- The need for distributed computing to parallelize the workload.
- High-dimensional data behave fundamentally differently (curse of dimensionality: sparsity, near-equal pairwise distances, near-orthogonality of vectors).

Sources referenced in the original lecture (for further reading): Han, Kamber, Pei — Data Mining: Concepts and Techniques (Morgan Kaufmann, 2012), ch. 10/11; Zaki & Meira — Data Mining and Machine Learning (2nd ed., Cambridge Univ. Press, 2020); Murphy — Probabilistic Machine Learning (MIT Press, 2022); James, Witten, Hastie, Tibshirani — An Introduction to Statistical Learning (Springer, 2021); Izenman — Modern Multivariate Statistical Techniques (Springer, 2008); Aggarwal — Data Mining: The Textbook; Kaufman & Rousseeuw (1990) — Finding Groups in Data; Ester et al. (1996) — DBSCAN (KDD'96); Leskovec, Rajaraman, Ullman — Mining of Massive Datasets.